

Fruit fly assay report

Session: /home/team8/z.avi_assay_tests/assayfinal/outputs/assay/assay_20260414_130947

Run dir: /home/team8/z.avi_assay_tests/assayfinal/outputs/assay/assay_20260414_130947

Raw video:

/home/team8/z.avi_assay_tests/assayfinal/outputs/assay/assay_20260414_130947/raw_video.mp4

Background:

/home/team8/z.avi_assay_tests/assayfinal/outputs/assay/assay_20260414_130947/background_transformed_snapshot.png

Calibration:

/home/team8/z.avi_assay_tests/assayfinal/outputs/assay/assay_20260414_130947/calibration_snapshot.json

Duration (s): 9.8

Frames processed: 50

Tracking mode: stitch

Mask blobs as flies: True

Assay tubes: 4

Track fragments detected: 20

Unique threshold crossings: 6

0 s to 9.8 s comparison window

0 s to 9.8 s x-displacement summary
Columns 1-6 of 9 | Rows 1-4 of 4

assay_tube_index	flies_compared	mean_abs_dx_px	median_abs_dx_px	mean_signed_dx_px	comparison_target_s
1	1	9.325	9.325	-9.325	9.8
2	4	9.837	5.871	-3.212	9.8
3	2	3.184	3.184	-3.029	9.8
4	3	39.209	45.722	-4.697	9.8

0 s to 9.8 s x-displacement summary
Columns 7-9 of 9 | Rows 1-4 of 4

mean_abs_dx_mm	median_abs_dx_mm	mean_signed_dx_mm
0.901	0.901	-0.901
1.041	0.621	-0.34
0.333	0.333	-0.317
4.285	4.997	-0.513

Threshold crossing summary
Columns 1-5 of 5 | Rows 1-4 of 4

assay_tube_index	number_of_flies_detected	number_of_unique_threshold_crossings	fraction_crossing_by_10s	first_crossing_time_s
1	1	0	0.0	nan
2	4	0	0.0	nan
3	5	0	0.0	nan
4	10	6	0.6	1.2

Threshold crossing events
Columns 1-6 of 8 | Rows 1-6 of 6

unique_event_id	assay_tube_index	display_id	crossing_time_s	crossing_x_px	crossing_y_px
1	4	9	1.2	875.258	226.84
2	4	10	1.8	970.79	239.938
3	4	5	3.0	876.083	238.075
4	4	6	5.0	969.428	250.699
5	4	1	6.2	972.069	254.966
6	4	3	8.4	925.458	244.932

Threshold crossing events
Columns 7-8 of 8 | Rows 1-6 of 6

[illegible]

Velocity summary
Columns 1-6 of 6 | Rows 1-4 of 4

assay_tube_index	number_of_flies_detected	mean_speed_mm_s	max_speed_mm_s	mean_vertical_velocity_mm_s	max_vertical_velocity_mm_s
1	1	0.253	0.849	-0.014	0.472
2	4	2.703	13.398	2.522	13.382
3	5	1.008	5.187	-0.05	2.587
4	10	8.072	24.204	6.824	23.781